

CS 105: Department Introductory Course on Discrete Structures

Instructor : S. Akshay

Sep 12, 2025

Lecture 12 – Basic structures: lattices, and then onto next
Module.

Logistics

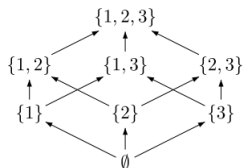
Midsem

- ▶ Saturday 20th
- ▶ More announcements on Piazza regarding seating and venue.
- ▶ Syllabus: all till last lecture including posets, chains, Mirsky's Theorem. No lattices.

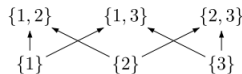
Optional basics help session

- ▶ Tomorrow, Saturday 3pm
- ▶ Last organized help session before midsem

Minimal and maximal elements



Poset $P_1 = (S_1, \subseteq)$
where $S_1 = \mathcal{P}(\{1, 2, 3\})$



Poset $P_2 = (S_2, \subseteq)$
where $S_2 = \mathcal{P}(\{1, 2, 3\}) \setminus \{\emptyset, \{1, 2, 3\}\}$

Let $(S, \preceq)$ be a poset.

- ▶ $a \in S$ is **minimal** in S if $\forall b \in S, b \preceq a \implies b = a$
- ▶ $a \in S$ is **maximal** in S if $\forall b \in S, a \preceq b \implies a = b$.
- ▶ $a \in S$ is **least/minimum** element of S if $\forall b \in S, a \preceq b$
- ▶ $a \in S$ is **greatest/maximum** element of S if $\forall b \in S, b \preceq a$.

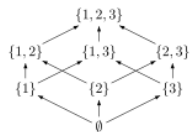
Examples: $\emptyset$ is a minimal and the minimum element in P_1 , $\{1\}, \{2\}, \{3\}$ are all minimal elements in P_2 , but P_2 does not have any minimum element.

Exercise: What are the maximal/maximum elements in P_1, P_2 ?

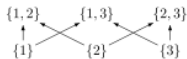
Least upper bounds and greatest lower bounds

Let $(S, \preceq)$ be a poset and $A \subseteq S$.

- ▶ $u \in S$ (resp. $l \in S$) is called an **upper bound** (resp. **lower bound**) of A iff $a \preceq u$ (resp. $l \preceq a$) for all $a \in A$.
- ▶ $u \in S$ is the **least upper bound (lub)** of A if it is an upper bound of A and is less than every other upper bound.
- ▶ $l \in S$ is the **greatest lower bound (glb)** of A if it is a lower bound of A and is greater than every other lower bound.



Poset $P_1 = (S_1, \subseteq)$
where $S_1 = \mathcal{P}(\{1, 2, 3\})$



Poset $P_2 = (S_2, \subseteq)$
where $S_2 = \mathcal{P}(\{1, 2, 3\}) \setminus \{\emptyset, \{1, 2, 3\}\}$

- ▶ Let $A = \{\{1\}, \{2\}\}$. Then $\{1, 2\}, \{1, 2, 3\}$ are upper bounds of A in P_1 and $\{1, 2\}$ is the lub of A . In P_2 , is there an lub for S_2 ?

Least upper bounds and greatest lower bounds

Let $(S, \preceq)$ be a poset and $A \subseteq S$.

- ▶ $u \in S$ (resp. $l \in S$) is called an **upper bound** (resp. **lower bound**) of A iff $a \preceq u$ (resp. $l \preceq a$) for all $a \in A$.
- ▶ $u \in S$ is the **least upper bound (lub)** of A if it is an upper bound of A and is less than every other upper bound.
- ▶ $l \in S$ is the **greatest lower bound (glb)** of A if it is a lower bound of A and is greater than every other lower bound.

Some Observations (**Exercise:** Prove it!)

- ▶ The lub/glb of a subset A in S , if it exists, is unique.
- ▶ If the lub/glb of $A \subseteq S$ belongs to A , then it is the greatest/least element of A .

Lattices

Definition

- ▶ A **lattice** is a poset in which every pair of elements has both a lub and a glb (in the set), i.e., $\forall x, y \in S$, there exists $l, u \in S$ such that l is the glb and u is the lub of $\{x, y\}$.
- ▶ $(\mathcal{P}(S), \subseteq)$ is a lattice.
- ▶ What about $(\{2, 4, 5, 10, 12, 20, 25\}, |)$, $(\{1, 2, 4, 8, 16\}, |)$, $(\mathbb{Z}^+, |)$?

Applications of Lattices

- ▶ Models of information flow – think security clearance.
- ▶ Finite lattices have a strong link with Boolean Algebra
- ▶ Several other applications in many domains of mathematics and CS, including **formal semantics of programming languages**, **program verification**.

Summary till now

Course Outline

1. Proofs and reasoning

- ▶ Propositions, predicates
- ▶ Proofs and proof techniques: contradiction, contrapositive, (strong) induction, well-ordering principle, diagonalization.

2. Basic discrete structures

- ▶ (finite and infinite) sets
- ▶ Functions: injections, surjections, bijections
- ▶ Relations: equivalence relations, partial orders, lattices
- ▶ Some applications

3. Counting and combinatorics

4. Introduction to graph theory

Next chapter: Counting and Combinatorics

Topics to be covered

- ▶ Basics of counting
- ▶ Subsets, partitions, Permutations and combinations
- ▶ Pigeonhole Principle and its extensions
- ▶ Recurrence relations and generating functions

Introduction to combinatorics

Does it really need an introduction

- ▶ Enumerative combinatorics: counting combinatorial/discrete objects e.g., sets, numbers, structures...
- ▶ Existential combinatorics: show that there exist some combinatorial “configurations”.
- ▶ Constructive combinatorics: construct interesting configurations...

Simple examples and principles...

- ▶ How many reflexive relations are there on a set A of size n ?
 - ▶ Reflexive relations are ordered pairs of which there are n^2 .
 - ▶ Of these, all n pairs of (a, a) have to be present.
 - ▶ Of the remaining, we can choose any of them to be in or out.
 - ▶ there are $n^2 - n$ of them, so $2^{n^2 - n}$ of them.
 - ▶ We used the so-called “product principle”...

The product principle

If there are n_1 ways of doing something and n_2 ways of doing another thing, then there are $n_1 \cdot n_2$ ways of performing both actions.

Simple examples and principles...

- ▶ How many reflexive relations are there on a set A of size n ?
- ▶ How many functions are there from a set of size n to itself?
- ▶ How many subsets does a set A of n elements have?
 - ▶ **Product principle**: two choices for each element, hence $2 \cdot 2 \cdots 2 \cdot 2$ (n -times).
 - ▶ **Bijection**: between $\mathcal{P}(X)$ and n -length sequences over $\{0, 1\}$ (characteristic vector).
 - ▶ **Induction**: Since we already know the answer!
 - ▶ **Recurrence**: $F(n) = 2 \cdot F(n-1)$, $F(0) = 1$. solve it?
 - ▶ **Sum principle**: Subsets of size 0 + subsets of size 1 + ... + subsets of size n = Total number of subsets.

Sum Principle

If something can be done in n_1 **or** n_2 ways such that none of the n_1 ways is the same as any of the n_2 ways, then the total number of ways to do this is $n_1 + n_2$.

Simple examples and principles...

- ▶ How many reflexive relations are there on a set A of size n ?
- ▶ How many functions are there from a set of size n to itself?
- ▶ How many subsets does a set A of n elements have?
 - ▶ **Product principle**: two choices for each element, hence $2 \cdot 2 \cdots 2 \cdot 2$ (n -times).
 - ▶ **Bijection**: between $\mathcal{P}(X)$ and n -length sequences over $\{0, 1\}$ (characteristic vector).
 - ▶ **Induction**: Since we already know the answer!
 - ▶ **Recurrence**: $F(n) = 2 \cdot F(n-1)$, $F(0) = 1$. solve it?
 - ▶ **Sum principle**: Subsets of size 0 + subsets of size 1 + ... + subsets of size n = Total number of subsets.
- ▶ But, how many subsets of size k does a set of n elements have? This number, denoted $\binom{n}{k}$, is called a **binomial coefficient**.
- ▶ We all know(?) that $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. Prove it!

Permutations and combinations

Binomial Coefficients. Let n, k be integers s.t., $n \geq k \geq 0$.

How many subsets of size k does a set of n elements have? This number, denoted $\binom{n}{k}$, is called a **binomial coefficient**.

One proof of $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is as follows: Let us count the number of “**ordered**” subsets of size k .

- ▶ No. of **ordered** subsets of size $k = n \cdot (n-1) \cdots (n-k+1)$.
- ▶ No. of **ordered** subsets of size $k = (\text{no. of unordered subsets}) \times (\text{no. of ways to order them}) = \binom{n}{k} \times k!$.
- ▶ Equate them! Principle of **double counting**.

Permutations and combinations

- ▶ No. of k -size subsets of set of size $n = \text{No. of } k\text{-combinations of a set of } n \text{ (distinct) elements} = \binom{n}{k}$.
- ▶ No. of k -size ordered subsets of set of size $n = \text{No. of } k\text{-permutations of a set of } n \text{ (distinct) elements}$.

Simple examples to illustrate “double counting”

Prove the following identities (by only using double counting!)

1. $\sum_{k=0}^n \binom{n}{k} = 2^n.$

2. $\binom{n}{k} = \binom{n}{n-k}.$

3. $k \binom{n}{k} = n \binom{n-1}{k-1}$

4. $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

The latter two are in fact recursive definitions for $\binom{n}{k}$. What are the boundary conditions?

A more interesting example with double counting

Handshake Lemma

At a meeting with n people, the number of people who shake hands an odd number of times is even.

What will you count here?